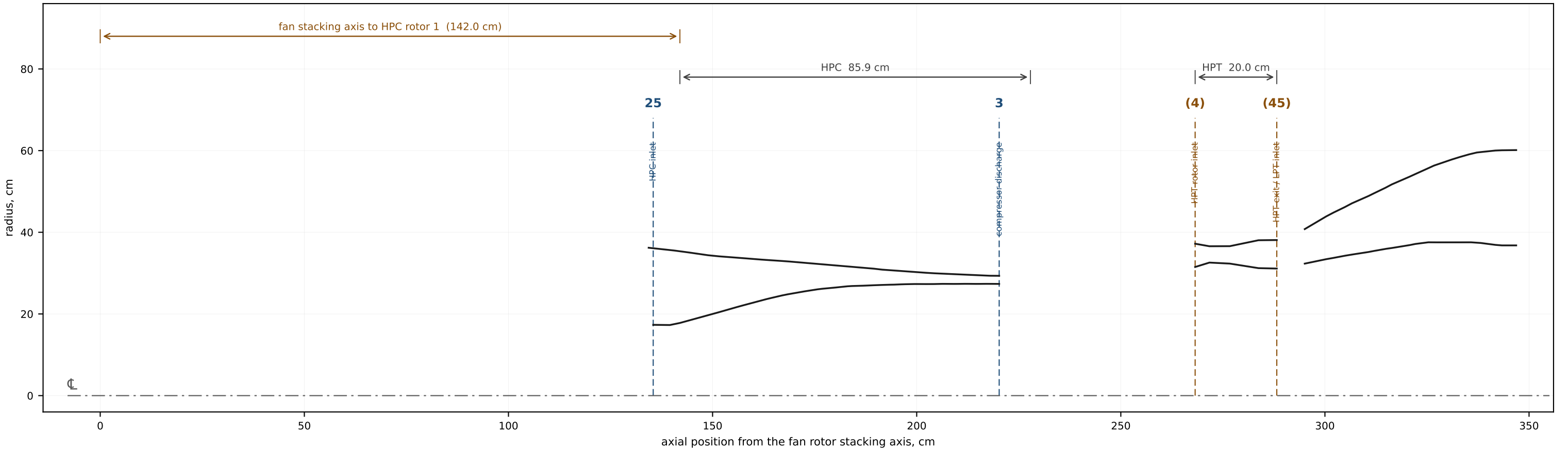




## GENERAL ARRANGEMENT — meridional half-section

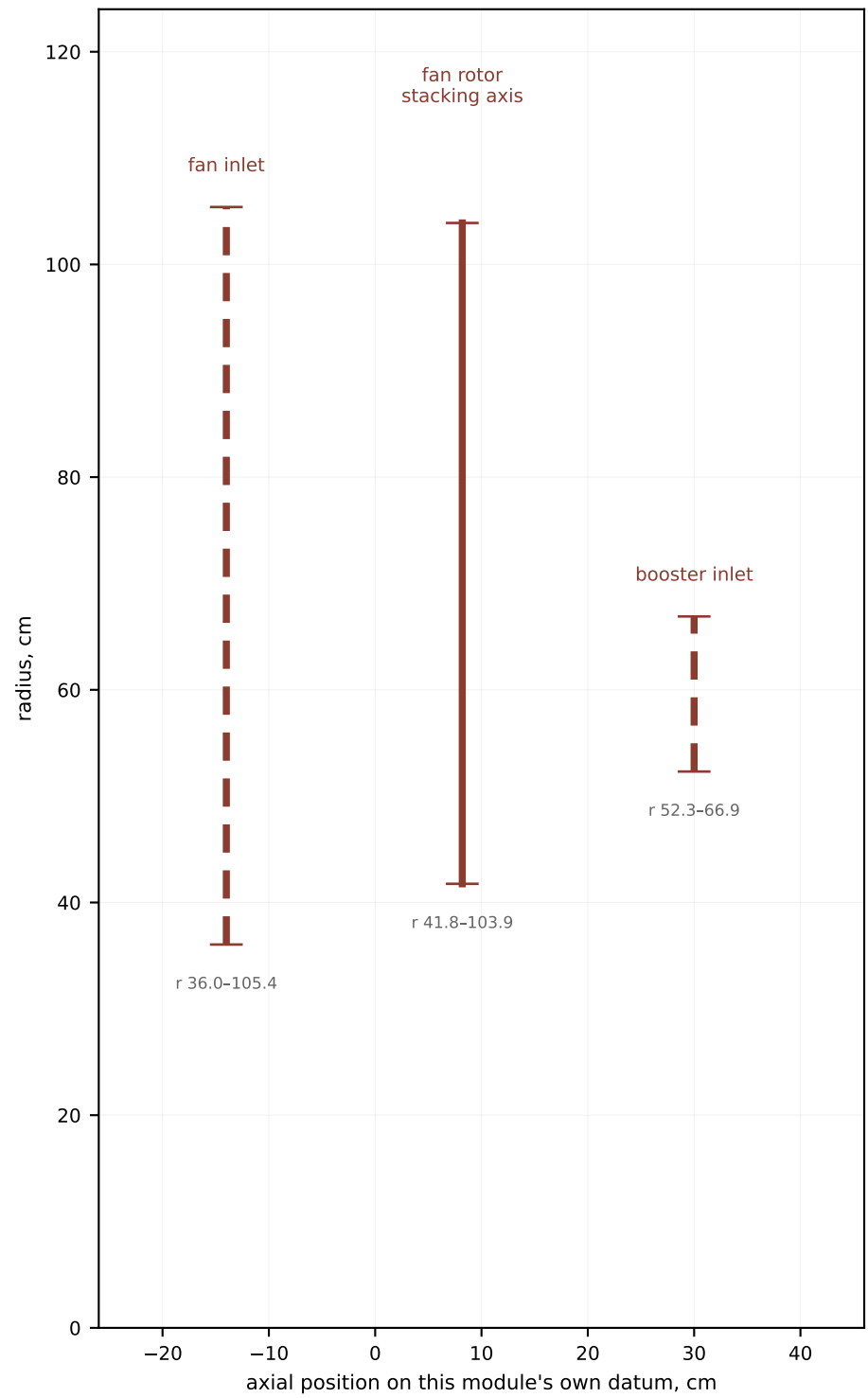
NASA/GE Energy Efficient Engine — Flight Propulsion System. Reconstructed from public-domain NASA reports.

STATIONS DRAWN: 25, 3, 4, 45. NOT DRAWN, no published axial position: 1 (freestream; no plane in this projection); 2 (upstream of the fan stacking axis by an unpublished amount) ...

## PF-09 E<sup>3</sup> RECONSTRUCTION

SHEET 1 OF 6

DATUM fan rotor stacking axis, x = 0  
UNITS centimetres  
PROJ meridional half-section  
( ) reference dimension — ASSUMED, not published



# FAN AND BOOSTER — radial stations

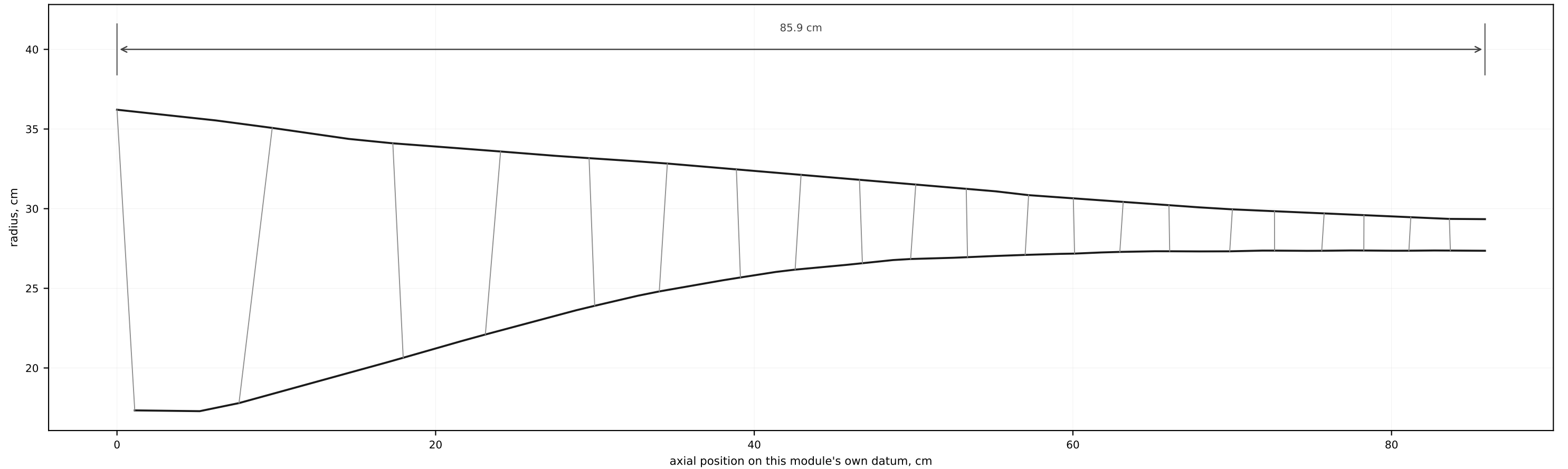
NASA/GE Energy Efficient Engine — Flight Propulsion System. Reconstructed from public-domain NASA reports.

NO WALL CONTOUR IS DRAWN. The fan module has three dimensioned radial stations and one axial position (Fig 15 stacking axis, solid). No hub or casing line was published against an axial coordinate, so none is drawn. Dashed bars are radii only, placed indicatively.

## PF-09 E<sup>3</sup> RECONSTRUCTION

SHEET 2 OF 6

DATUM fan rotor stacking axis, x = 0  
 UNITS centimetres  
 PROJ meridional half-section  
 ( ) reference dimension — ASSUMED, not published



## HIGH-PRESSURE COMPRESSOR — 10 stages, 21 rows

NASA/GE Energy Efficient Engine — Flight Propulsion System. Reconstructed from public-domain NASA reports.

Hub and tip from Table XXI streamlines 1 and 12, 42 stations. Row lines mark leading edges. Pressure ratio 23.0 (CR-168219 Table XIV); the HPC report's abstract quotes 22.6 and both are recorded.

## PF-09 E<sup>3</sup> RECONSTRUCTION

SHEET 3 OF 6

DATUM fan rotor stacking axis,  $x = 0$   
UNITS centimetres  
PROJ meridional half-section  
( ) reference dimension — ASSUMED, not published

generated by solvers/publication/drawings.py

# NO DIMENSIONED GEOMETRY PUBLISHED

The double-annular combustor appears in Figures 1, 22 and 79 as undimensioned drawings.  
Liner axial coordinates, dome height and the liner hole areas are not printed anywhere in the reports read.

What IS published: double-annular, 60 cups, 30 fuel nozzles (CR-168219 sec 5.3 p.57),  
and the exit temperature profile that Stage D's cooling design is built on.

This sheet is deliberately empty. An empty sheet with the reason is a result;  
a missing sheet is a silence.

**COMBUSTOR — double annular**

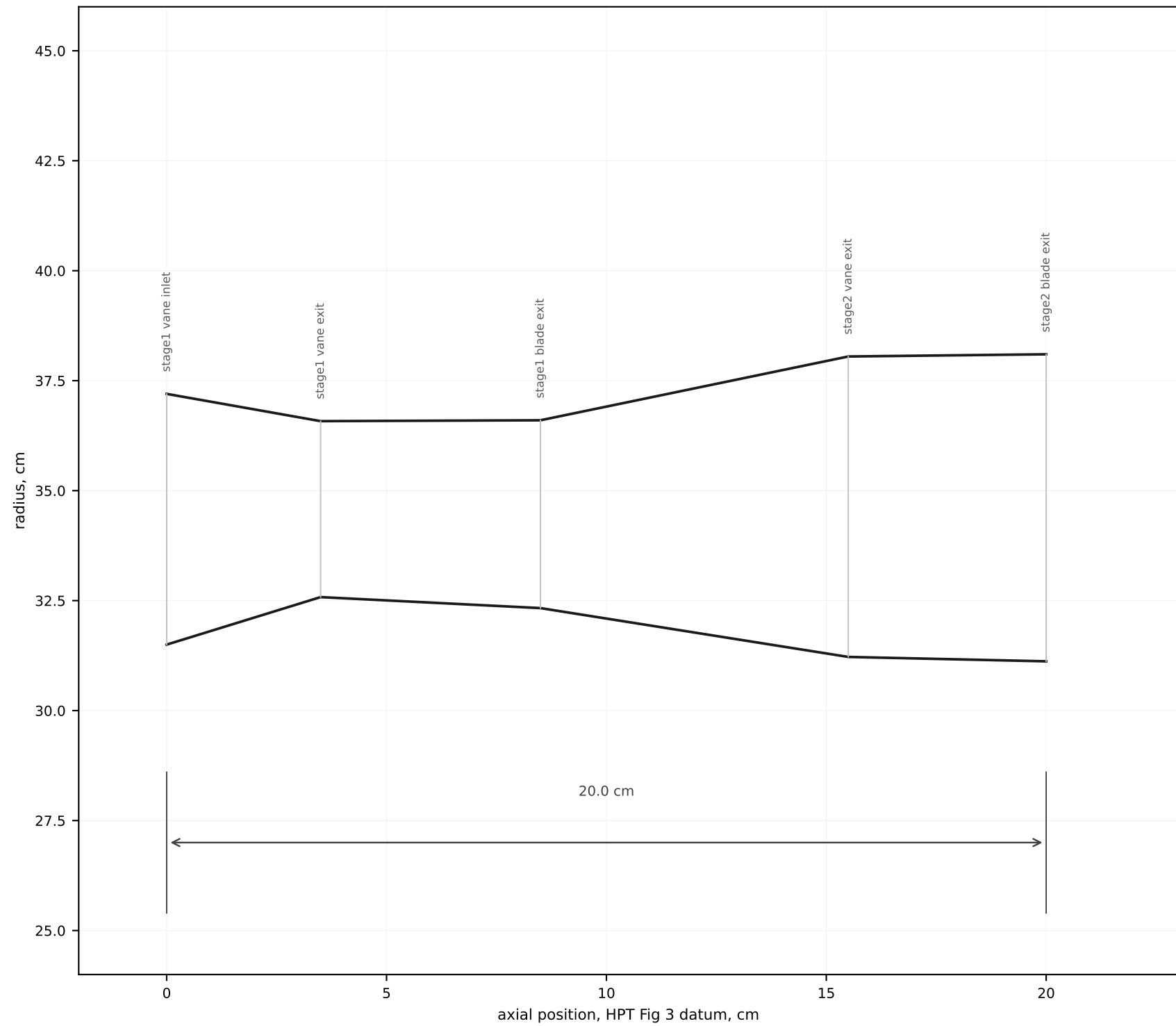
NASA/GE Energy Efficient Engine — Flight Propulsion System. Reconstructed from public-domain NASA reports.

A3 backlog item. It also gates unit D2 (liner hole areas) and is half the reason the HPC-to-HPT axial offset must be assumed rather than built up.

**PF-09 E<sup>3</sup> RECONSTRUCTION**

SHEET 4 OF 6

DATUM fan rotor stacking axis, x = 0  
UNITS centimetres  
PROJ meridional half-section  
( ) reference dimension — ASSUMED, not published



## HIGH-PRESSURE TURBINE — 2 stages

NASA/GE Energy Efficient Engine — Flight Propulsion System. Reconstructed from public-domain NASA reports.

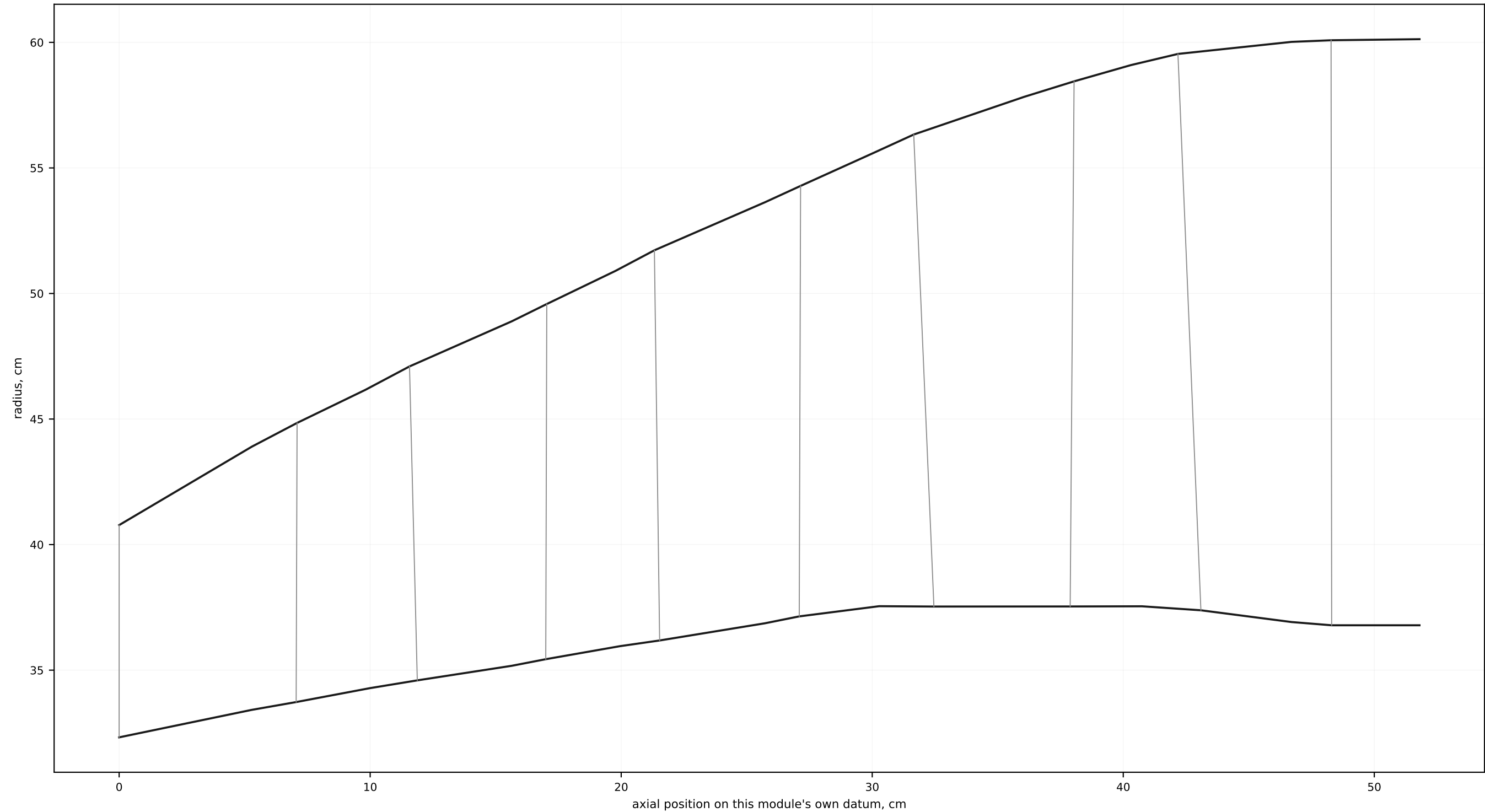
Five dimensioned stations from HPT report Fig 3. NO AIRFOIL IS DRAWN: the HPT blade and vane coordinates were never published — the reports give throat dimensions and aspect ratio only. Radii are printed; their axial positions are read off the drawing's own scale,  $\pm 0.3$  cm.

## PF-09 E<sup>3</sup> RECONSTRUCTION

SHEET 5 OF 6

DATUM fan rotor stacking axis,  $x = 0$   
 UNITS centimetres  
 PROJ meridional half-section  
 ( ) reference dimension — ASSUMED, not published

generated by solvers/publication/drawings.py



# LOW-PRESSURE TURBINE — 5 stages, 10 rows

NASA/GE Energy Efficient Engine — Flight Propulsion System. Reconstructed from public-domain NASA reports.

Walls and rows from the thirty printed section coordinates. This module's z runs from the HPT exit plane, which is the one inter-module offset the reports do give — the transition duct at the left is measured, not assumed.

## PF-09 E<sup>3</sup> RECONSTRUCTION

SHEET 6 OF 6

DATUM fan rotor stacking axis, x = 0  
UNITS centimetres  
PROJ meridional half-section  
( ) reference dimension — ASSUMED, not published